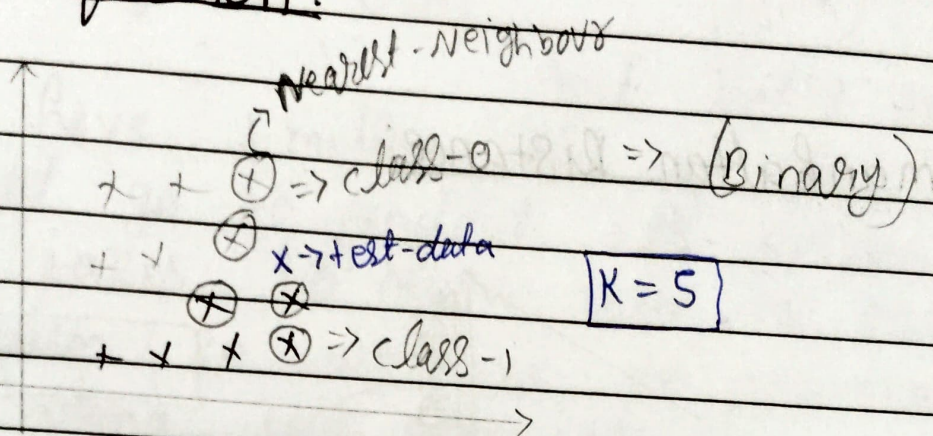


K - Nearest Neighbour (KNN)

- (i) Classification
- (ii) Regression

(i) Classification:



(i) $\Rightarrow$ we have to initialize the K-value.

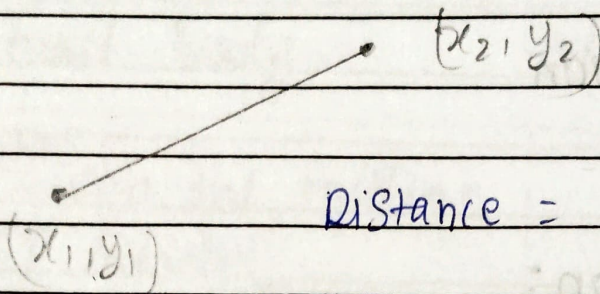
$K > 0$ and it is a Hyperparameter

$K = \{1, 2, 3, 4, \dots, \infty\} \Rightarrow$ Let $K=5$

(ii) Find the K-nearest neighbour for the Test data

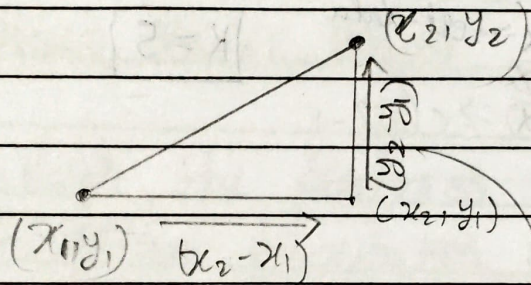
(iii) From those $K=5$ how many neighbour belong to class-0 and class-1

① Euclidian distance



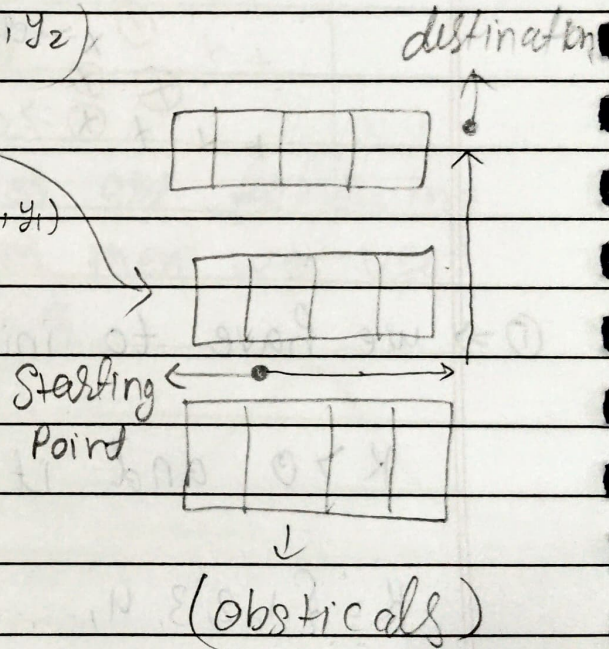
$$\text{Distance} = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

② manhattan distance:

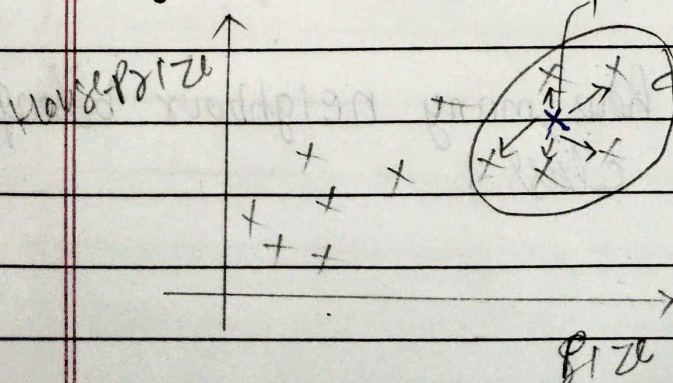


Distance ↓

$$|x_2 - x_1| + |y_2 - y_1|$$



③ Regression.



avg of all the data points for the output.

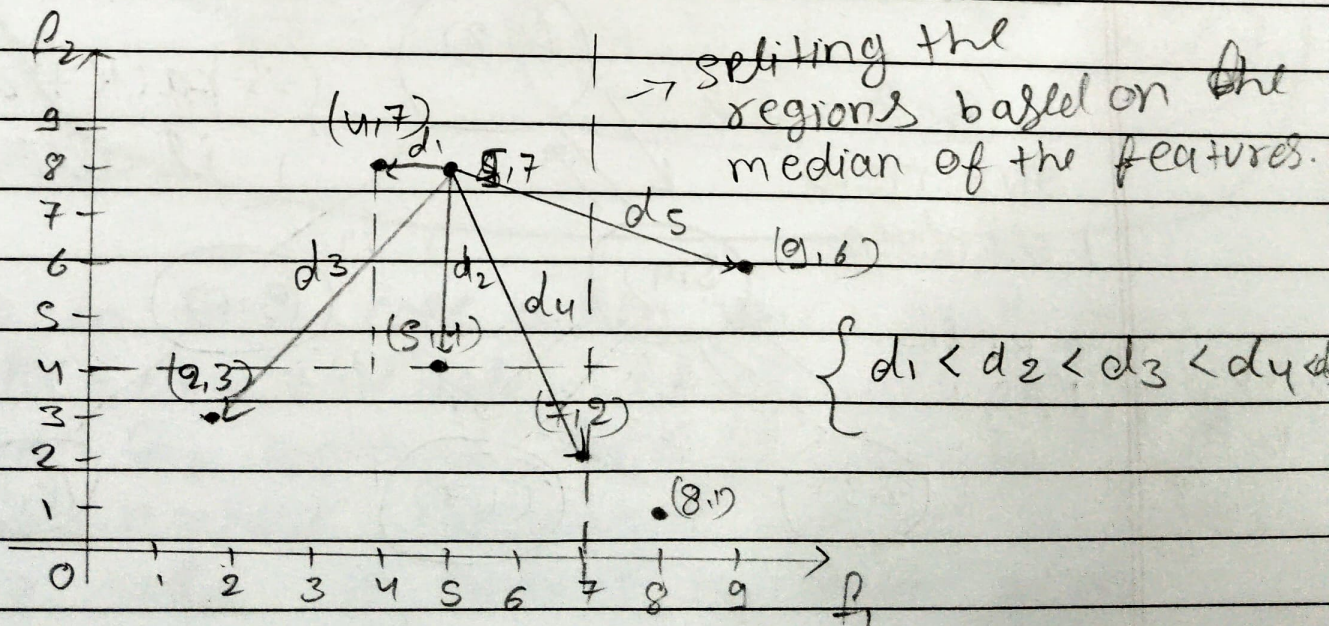
So we are understand the concept of KNN that how it is work, but the problem is Time why?

if It is calculate the Distance from the test data that's why its time complexity is $O(n)$,

but Let's suppose you have 1 million data if your KNN model go to findout it's distance it takes to much time to solve you problem, so we have to optimized it by using it's variants.

Optimization of KNN - KDtree and Ball tree

① KDtree:



P_1	P_2
2	3
4	7
5	4
7	2
8	1
9	6

median of $P_2 =$

1, 2, 3, 4, 6, 7

$$\frac{3+4}{2} = \frac{7}{2} = 3.5$$

you take either
3 or 4

$$\text{median} = \frac{8+7}{2}$$

$$= 6.5$$

↓
we select
the 7
from the

P_1 to divide

let it be a 4

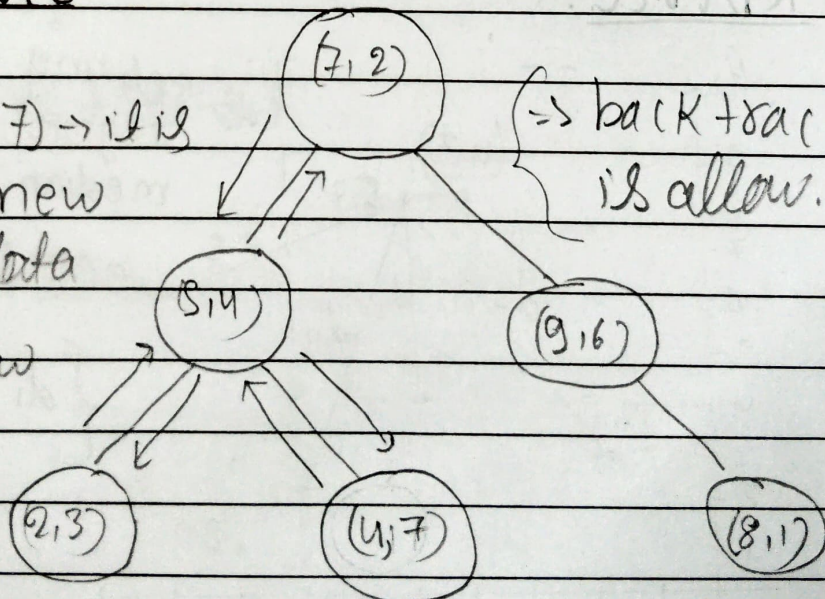
and in P_2 draw the
line passing to 4
and divided in
regions.

KD Tree

$(5, 7) \rightarrow$ it is
our new
test data

$\Rightarrow$ backtracking
is allow.

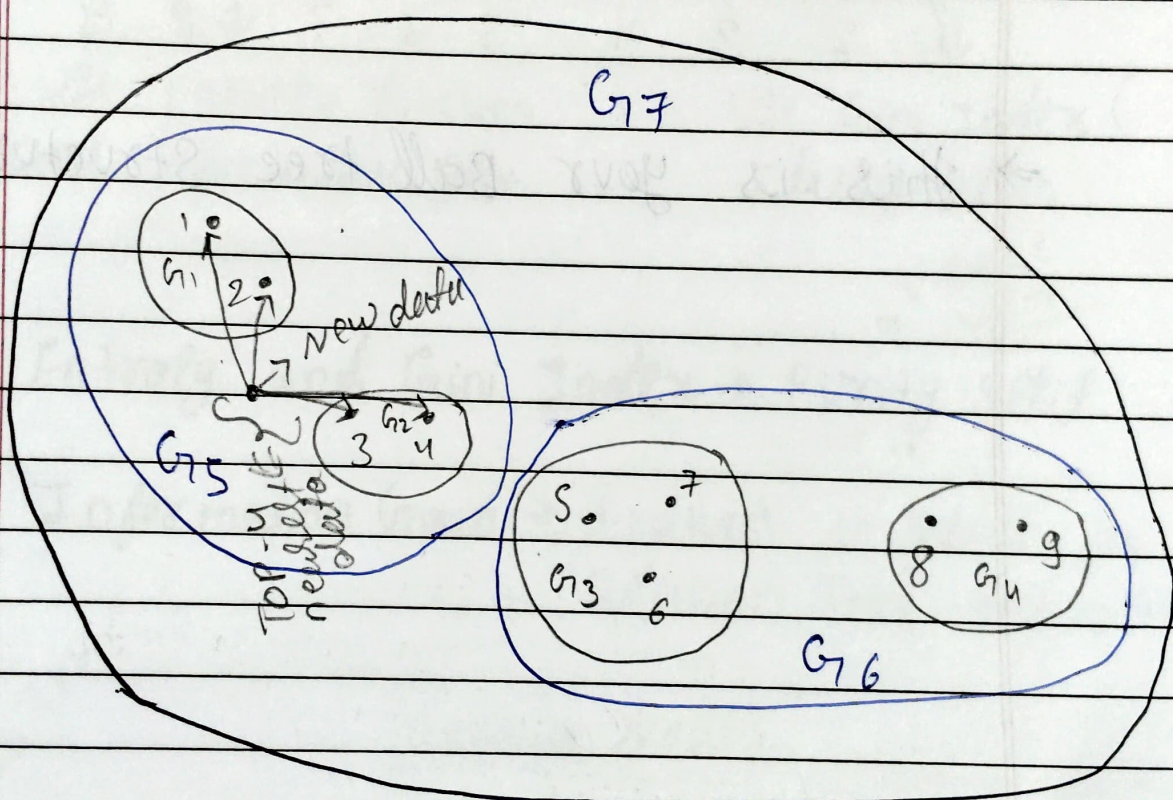
$\Rightarrow$ Now we know
where it is
fall.



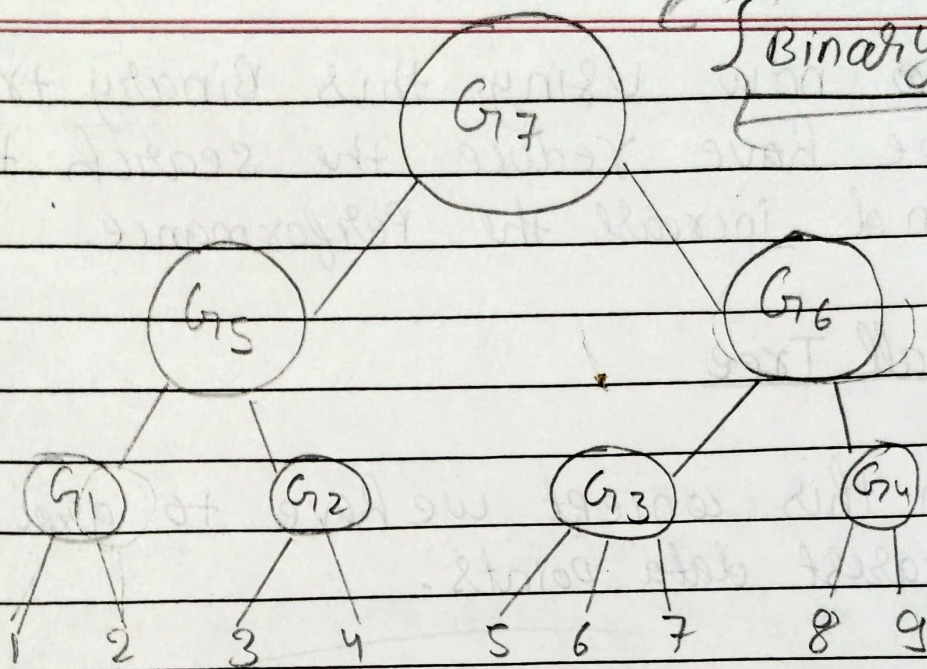
So now using this Binary tree we have reduce the search time and increase the Performance.

② Ball Tree

In this concept we have to group the nearest data points.



So in this case when you check your data (new test data) it lies in G_5 and G_5 contains G_2 & G_1 . So your Top nearest neighbors belongs to G_1 & G_2 no need to search on other Groups.



⇒ This is your Ball tree structure